ORCHESTRA – a briefing note for AAGI and GRDC

An R-package federation for decisions that know when to refuse

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Decision in one line

An open, typed federation of R packages now exists that couples a crop model, its inversion against field data, a causal test of whether the field agrees with the model, a decision layer that refuses to price a rejected posterior, and a value-of-information map of where the next trial or soil test pays. On a controlled synthetic barley belt it cuts realised loss against a model-only recommendation by 49 per cent (2,960 vs 5,818 AUD/ha summed), reproduced on a re-seeded belt. The ask is a conversation about running the same chain on a real trial archive.

What it is

Sixteen member packages (seven more as candidates) developed under one contract: every inference result is an S7 `orchestra_manifest` with a payload hash and lineage, and every refusal or abstention is a typed object that any downstream member reads (`is_orchestra_decline()`). Members include APSIM Next Gen as a member (apsimR), simulator inversion that drives real PEST++ (PESTO), kernR's mechanism-consistency test, a multi-backend Bayesian hub (flexyBayes), a loss-and-value-of-information decision layer (decideR), a grower-facing planner that also refuses (grainPlan), and sovereign synthetic twins of confidential trial data (masque). All are MIT or GPL licensed and published or staged under `max578` with an R-universe channel; no GRDC-origin code or data is inside any package.

What Study A0 showed

- **Set-up.** 40 sites on real grain-belt coordinates with real climate and soil, five seasons, six nitrogen rates, three varieties, three replicates, yields from APSIM. At 12 sites a hidden subsoil constraint flattens the response above 60 kg N/ha; the model used for inversion is never told.
- **Detection.** The mechanism-consistency test refuses the model where the field disagrees (sensitivity 0.58, specificity 0.93); the field record, read through a plateau-capable lens, places the plateau onset's posterior interval over the planted 60 kg N/ha at 12 of 12 constrained sites.
- **Value.** Cascade loss 2,960 vs naive 5,818 AUD/ha summed over 198 site-seasons; at constrained cells 27.0 vs 75.2 per site-season; 1,440 kg N not applied and 6.2 t CO_2-e/ha avoided. Re-seeded belt: 2,489 vs 5,175.
- **Where to look next.** An expected-value-of-sample-information map prices one strip trial or one soil test at 40 of 40 sites (top site 164 AUD/ha).
- **Failures on the record.** Seven independent oracles run inside the study scripts; two fail and are reported as failures (a plateau-onset separation test and an under-powered site-level test). Version 0.1 of the study lost to the naive arm because its second lens could not express a plateau; the finding was reported before the lens was changed.

What it does not yet show

A synthetic belt, not a real one: the shape conclusions have real-data cover (the belt partially brackets a real barley multi-environment trial's N-response slopes), the dollar magnitudes do not. The value-of-information figures are directional. The information prices, the district status-quo rate and the carbon price are inputs awaiting validation, stated as such.

What we ask

1. **A real trial archive** to run the same chain on, under the sovereign-twin route if the data cannot leave its owner (masque produces a faithful synthetic clone; analysis is developed on the clone and replayed on the real data by its custodian).
2. **The AAGI-AUS mirror.** The site and the packages are staged under `max578`; mirroring under AAGI-AUS is ready once the licence conversation with the GRDC manager has happened. No licence is declared for the mirror until then.
3. **A slot to demonstrate.** The end-to-end walk on one site runs offline in under a minute (`demo/a0_demo.R`); the whole study re-runs from a driver.

Evidence. `max578.github.io/orchestra` (site, member atlas, contract). `studies/A0/REPORT.md` (every number bound to an object), `SUMMARY.md`, seven oracle files and a re-seeded re-verification. Fleet refusal gate `integration/test_refusal_contract.R` (20 of 20, 2026-08-26). Version freeze `integration/orchestra_full_ride/run_card_showcase_freeze_2026-08-26.md`.